Advanced Distributed Systems Development with Multiagent Systems

Communication as action

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Communication

Communication, in general, is the intentional exchange of information brought about by the production and perception of signs drawn from a shared system of conventional signs.

"Classical" view (pre-1953):

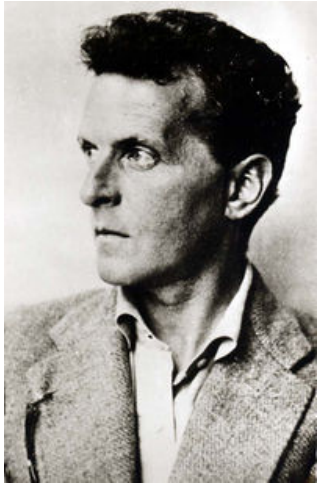
language consists of sentences that are true/false (cf. logic)

"Modern" view (post-1953):

language is a form of action

Communication

Wittgenstein (1953) **Philosophical Investigations**

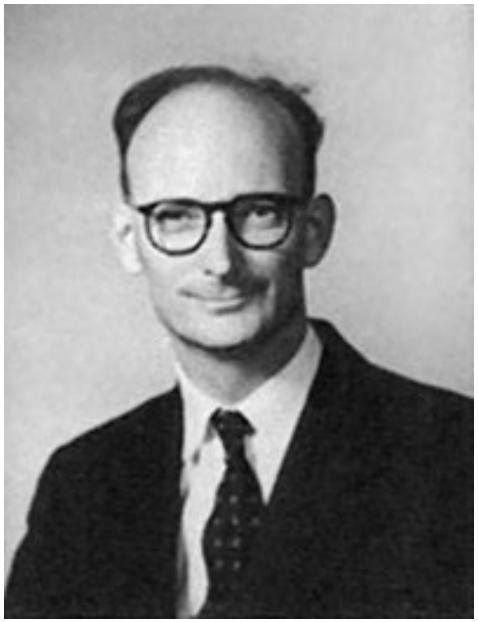


According to Wittgenstein,

- The structure of reality determines the structure of language.
- Meaning is determined by use, in the context of a "language-game".
- Words are not defined by reference to the objects or things which they designate in the external world nor by the thoughts, ideas, or mental representations that one might associate with them, but rather by how they are used in effective, ordinary communication.

Communication

Austin (1962) **How to Do Things with Words**



- In saying something, we do something, such as when a minister joins two people in marriage saying "I now pronounce you husband and wife".

Communication

Searle (1969) **Speech Act Theory**



- Each illocution can be described in terms of what it is attempting to do. So an assertion counts as an undertaking that something really is the case. A question counts as an attempt to elicit some information. Thanking someone counts as an expression of gratitude.
- Assumed intent of the speaker, or the *intentionality* of the sentence.

Speech acts



Speech acts achieve the speaker's goals:

Inform	"There's a pit in front of you"
Query	"Can you see the gold?"
Command	"Pick it up"
Promise	"I'll share the gold with you"
Acknowledge	"OK"

Speech act planning requires knowledge of

- Situation
- Semantic and syntactic conventions
- Hearer's goals, knowledge base, and rationality

Stages in communication

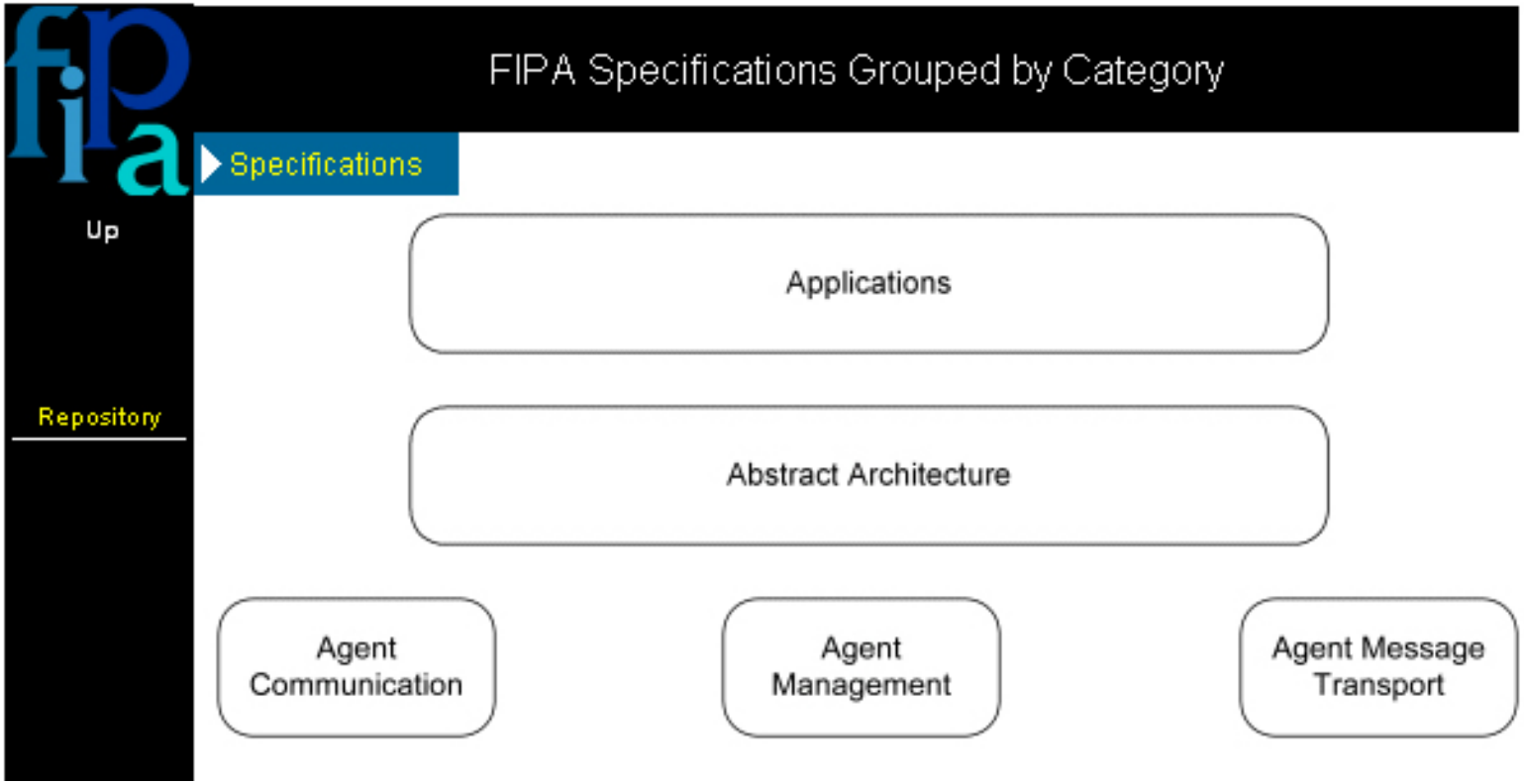
Intention	S wants to inform H that P
Generation	S selects words W to express P in context C
Synthesis	S utters words W
Perception	H perceives W' in context C'
Analysis	H infers possible meanings $P_1, \ldots P_n$
Disambiguation	H infers intended meaning P_i
Incorporation	H incorporates P_i into KB

Stages in communication

How could this go wrong?

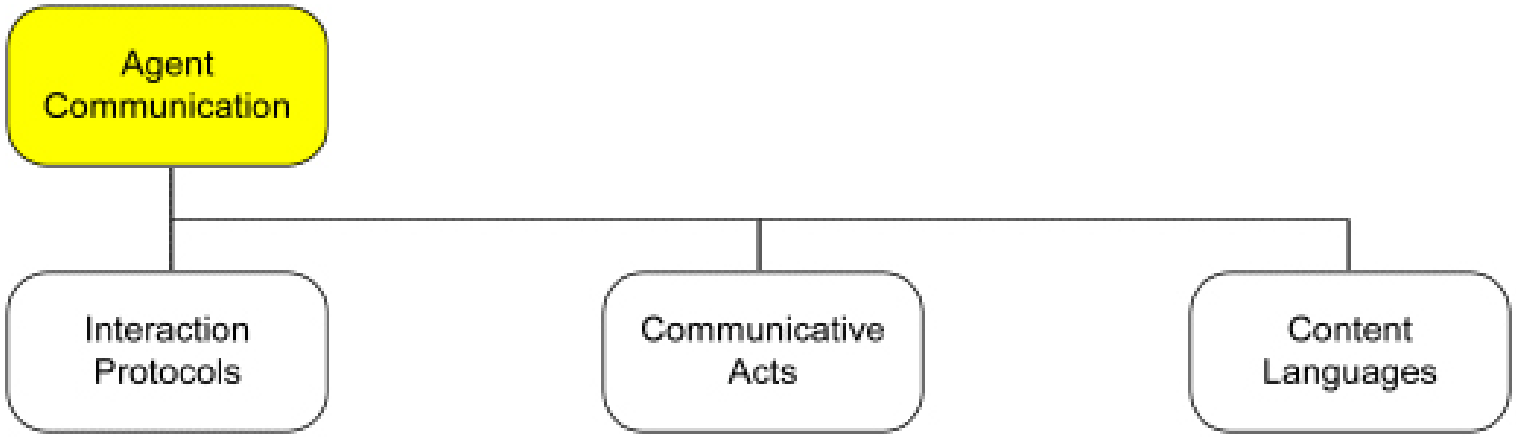
- Insincerity (S doesn't believe P)
- Speech wreck ignition failure
- Ambiguous utterance
- Differing understanding of current context ($C \neq C'$)

FIPA specifications: an overview



`http://www.fipa.org/repository/bysubject.html`

FIPA specifications: communication



`http://www.fipa.org/repository/aclspecs.html`

FIPA Message Structure

Parameter	Category of Parameters
performative	Type of communicative acts
sender	Participant in communication
receiver	Participant in communication
reply-to	Participant in communication
content	Content of message
language	Description of Content
encoding	Description of Content
ontology	Description of Content
protocol	Control of conversation
conversation-id	Control of conversation
reply-with	Control of conversation
in-reply-to	Control of conversation
reply-by	Control of conversation

http://www.fipa.org/specs/fipa00061/SC00061G.pdf

FIPA Communicative Acts

- Accept Proposal
- Agree
- Cancel
- Call for Proposal
- Confirm
- Disconfirm
- Failure
- Inform
- Inform If
- Inform Ref
- Not Understood
- Propagate
- Propose
- Proxy

FIPA Communicative Acts

- Query If
- Query Ref
- Refuse
- Reject Proposal
- Request
- Request When
- Request Whenever
- Subscribe

`http://www.fipa.org/specs/fipa00037/SC00037J.pdf`

Accept proposal summary

Accept proposal: The action of accepting a previously submitted proposal to perform an action.

Accept proposal content

A tuple consisting of an action expression denoting the action to be done, and a proposition giving the conditions of the agreement.

Accept proposal description

`accept-proposal` is a general-purpose acceptance of a proposal that was previously submitted (typically through a `propose` act). The agent sending the acceptance informs the receiver that it intends that (at some point in the future) the receiving agent will perform the action, once the given precondition is, or becomes, true.

The proposition given as part of the acceptance indicates the preconditions that the agent is attaching to the acceptance. A typical use of this is to finalize the details of a deal in some protocol. For example, a previous offer to "hold a meeting anytime on Tuesday" might be accepted with an additional condition that the time of the meeting is 11.00.

Note for future extension: an agent may intend that an action become done without necessarily intending the precondition. For example, during negotiation about a given task, the negotiating parties may not unequivocally intend their opening bids: agent *a* may bid a price *p* as a precondition, but be prepared to accept price p'.

Accept proposal formal model

$\langle i,$ accept-proposal $(j, \langle j, act\rangle, \phi)\rangle \equiv$

$\langle i,$ inform $(j, I_i$ Done $(\langle j, act\rangle, \phi))\rangle$

FP: $B_i\ \alpha \wedge \neg B_i\ (\text{Bif}_j\ \alpha \vee \text{Uif}_j\ \alpha)$

RE: $B_j\ \alpha$

Where:

$\alpha = I_i$ Done $(\langle j, act\rangle, \phi)$

Accept proposal example

Agent *i* informs *j* that it accepts an offer from *j* to stream a given multimedia title to channel 19 when the customer is ready. Agent *i* will inform *j* of this fact when appropriate.

```
(accept-proposal
  :sender (agent-identifier :name i)
  :receiver (set (agent-identifier :name j))
  :in-reply-to bid089
  :content
    "((action (agent-identifier :name j)
      (stream-content movie1234 19))
     (B (agent-identifier :name j)
      (ready customer78)))"
  :language fipa-sl)
```

FIPA Ontology Agent

- The model of agent communication in FIPA is based on the assumption that two agents, who wish to converse, share a common ontology for the domain of discourse.
- It ensures that the agents ascribe the same meaning to the symbols used in the message.
- For a given domain, designers may decide to use ontologies that are explicit, declaratively represented (and stored somewhere) or, alternatively, ontologies that are implicitly encoded with the actual software implementation of the agent themselves and thus are not formally published to an ontology service.

FIPA Ontology Agent

- The FIPA Ontology Service Specification (`http://www.fipa.org/specs/fipa00086/index.html`) deals with technologies enabling agents to manage explicit, declaratively represented ontologies.
- An ontology service for a community of agents is specified for this purpose. It is required that the service be provided by a dedicated agent, called an Ontology Agent (OA), whose role in the community is to provide some or all of the following services:

FIPA Ontology Agent

1. discovery of public ontologies in order to access them
2. maintain (for example, register with the DF, upload, download, and modify) a set of public ontologies,
3. translate expressions between different ontologies and/or different content languages,
4. respond to query for relationships between terms or between ontologies, and,
5. facilitate the identification of a shared ontology for communication between two agents.

FIPA Content Language

FIPA Content Language (CL) Specifications deal with different representations of the content of ALC messages. There are four languages that are suitable for being used as content languages in FIPA messages, although others can be added by the developer.

- SC00008 FIPA SL Content Language Specification
- XC00009 FIPA CCL Content Language Specification
- XC00010 FIPA KIF Content Language Specification
- XC00011 FIPA RDF Content Language Specification

`http://www.fipa.org/repository/cls.php3`

FIPA interaction protocols

FIPA Interaction Protocols (IPs) specifications deal with pre-agreed message exchange protocols for ACL messages.

- SC00026 FIPA Request Interaction Protocol Specification
- SC00027 FIPA Query Interaction Protocol Specification
- SC00028 FIPA Request When Interaction Protocol Specification
- SC00029 FIPA Contract Net Interaction Protocol Specification
- SC00030 FIPA Iterated Contract Net Interaction Protocol Specification
- XC00031 FIPA English Auction Interaction Protocol Specification
- XC00032 FIPA Dutch Auction Interaction Protocol Specification
- SC00033 FIPA Brokering Interaction Protocol Specification
- SC00034 FIPA Recruiting Interaction Protocol Specification
- SC00035 FIPA Subscribe Interaction Protocol Specification
- SC00036 FIPA Propose Interaction Protocol Specification

`http://www.fipa.org/repository/ips.php3`

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